Lecture 30
Chi-square tests

Review

1 In a matched-pairs t-test, what are the data?

the differences d within each pair: a one-sample t-test on d, with $df = n - 1$

2 Why is the paired SE usually smaller than the unpooled SE?

each pair shares a subject, so person-to-person spread cancels in d

3 What does the matched-pairs t-test assume?

the pairs are a random sample, and the differences are normal or $n \geq 30$

More than two categories?

So far a categorical response had two outcomes (yes or no), and we tested one p or $p_1 - p_2$. The Pew question has four answers, and someone claims all four are equally common among U.S. teens.

Never 289, Rarely 216, Sometimes 178, Often 60; $n = 743$

(a) What kind of variable is X? How many categories does it have?

(b) Can a test of one proportion p check the claim? Why or why not?

(a) categorical, with $k = 4$ categories

(b) no: one p covers one category; the claim is about all four, p_1, p_2, p_3, p_4, at once

Tables for categorical data

one-way table

the count in each category of one categorical variable

two-way (contingency) table

the count for every combination of two categorical variables: rows × columns

marginal distribution

one variable on its own: the row or column totals, as proportions

conditional distribution

one variable within a single row (or column) of the other, as proportions

Data: volunteers and hours

Type of volunteer	1–3 hours	4–6 hours	7–9 hours	Row total
Community college	111	96	48	255
Four-year college	96	133	61	290
Nonstudents	91	150	53	294
Column total	298	379	162	839

Adults 21 and older volunteer one to nine hours a week with a disabled senior citizen. A sample of 839 volunteers: X = type of volunteer, Y = hours volunteered per week. (OpenStax, Example 11.6)

Example: do the rows look alike?

Is the number of hours volunteered related to the type of volunteer? Before any test, compare the rows of the two-way table.

1–3 hours: community college 111 of 255; nonstudents 91 of 294; everyone 298 of 839

(a) What share of community college volunteers work 1–3 hours? Of nonstudents?

(b) What share of all 839 volunteers work 1–3 hours? Which kind of distribution is each?

(c) If hours had nothing to do with type, what would the rows look like?

(a) $111/255 = 0.435$ **and** $91/294 = 0.310$**: conditional distributions**

(b) $298/839 = 0.355$**: the marginal distribution of hours**

(c) every row would have about the same split as the column totals: 0.355, 0.452, 0.193

Observed and expected counts

observed count O

the count in a category (or cell) of our data

expected count E

the count we would expect if H_0 is true: $E = n\,p_i$, with p_i from H_0

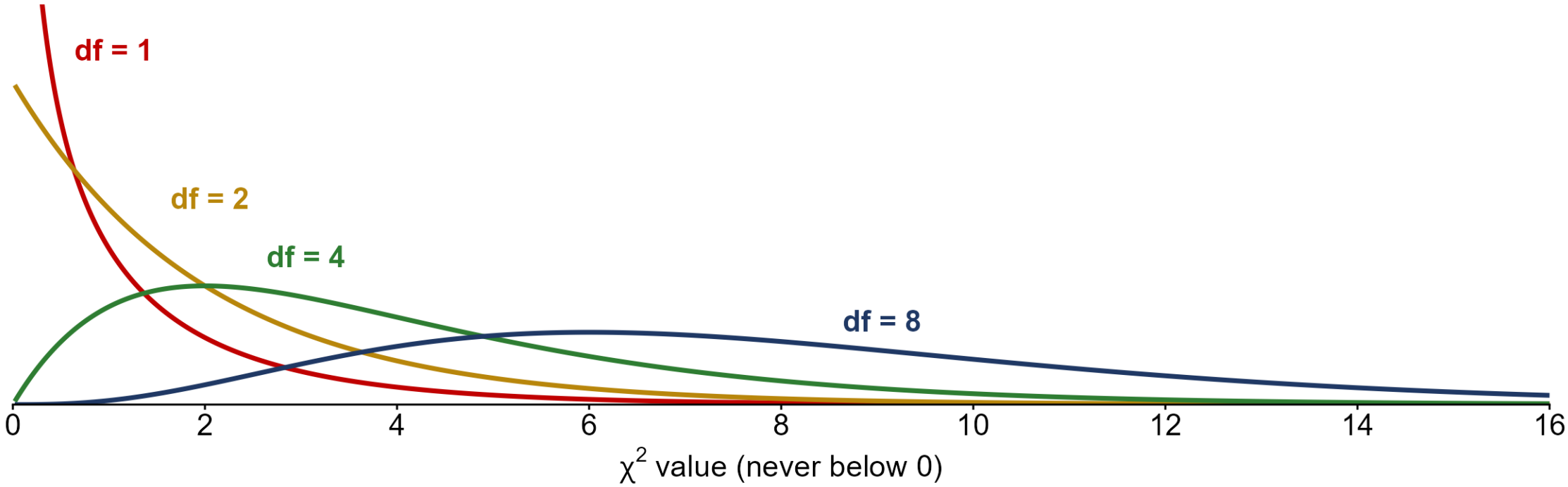
the chi-square distribution

a family of right-skewed curves for values ≥ 0, one for each df

degrees of freedom (one variable)

$df = k - 1$, **where k is the number of categories**

The chi-square distribution



never below 0, skewed right; mean $= df$; a larger df moves the curve right

The chi-square statistic

the chi-square statistic

$$X^2 = \sum \frac{(O - E)^2}{E}$$

added up over every category: how far each count is from what H_0 predicts

$X^2 = 0$

the observed counts match H_0 exactly

a large X^2

the counts are far from H_0: evidence against H_0 (always right-tailed)

if H_0 is true

X^2 is approximately $\chi^2(k-1)$, when every $E \geq 5$

Example: equally common answers?

Back to the Pew survey. Someone claims the four answers are equally common among U.S. teens:

$H_0: p_1 = p_2 = p_3 = p_4 = 0.25$

$n = 743$; **observed: Never 289, Rarely 216, Sometimes 178, Often 60**

(a) Find the expected count E for each answer.

(b) Find X^2.

(c) What is the df? Is X^2 large for that chi-square curve?

(a) $E = 743 \times 0.25 = 185.75$ **for every answer**

(b) $X^2 = 57.39 + 4.93 + 0.32 + 85.13 = 147.77$

(c) $df = 4 - 1 = 3$; **the mean of $\chi^2(3)$ is 3, so 147.77 is far out in the right tail**

The goodness-of-fit test

the question

does one categorical variable follow a claimed distribution $p_{1,0}, \ldots, p_{k,0}$**?**

the hypotheses

H_0**: every** $p_i = p_{i,0}$ **(it fits);** H_A**: at least one** $p_i \neq p_{i,0}$

the expected counts

$E_i = n\,p_{i,0}$**: what** H_0 **predicts for each category**

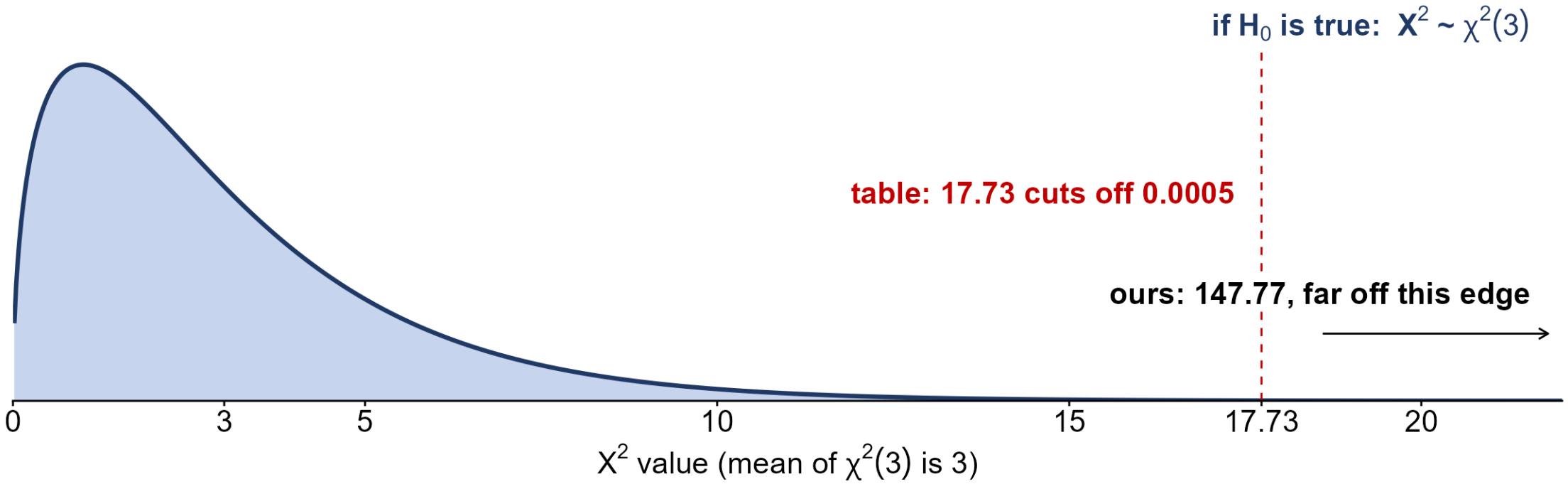
the statistic and df

$X^2_{obs} = \sum(O - E)^2/E$**, with** $df = k - 1$

the P-value

$P(\chi^2 \geq X^2_{obs} | H_0 \text{ true})$**: always the upper tail; needs every** $E \geq 5$

The Pew P-value



P-value < 0.0005 < $\alpha = 0.05$: reject H_0; the four answers are not equally common

Data: employee absences

Day	Mon	Tue	Wed	Thu	Fri	Total
Observed O	15	12	9	9	15	**60**

A random sample of 60 managers each named the weekday on which they had the most employee absences. (OpenStax, Example 11.2)

Example: equal absences?

Most employers believe employees are absent equally across a five-day week. A random sample of 60 managers named the day with the most absences. (OpenStax, Example 11.2)

Mon 15, Tue 12, Wed 9, Thu 9, Fri 15; $n = 60$, $\alpha = 0.05$

(a) State H_0 and H_A.
(b) Find the expected counts and X^2_{obs}.
(c) What is the df? The P-value is 0.5578. Decision?

(a) H_0: $p_1 = p_2 = p_3 = p_4 = p_5 = 0.2$; H_A: **at least one** $p_i \neq 0.2$
(b) $E = 60 \times 0.2 = 12$; $X^2_{obs} = 0.75 + 0 + 0.75 + 0.75 + 0.75 = 3$
(c) $df = 4$; **P-value = 0.5578 >** $\alpha = 0.05$: **fail to reject** H_0

Absences: expected counts and contributions

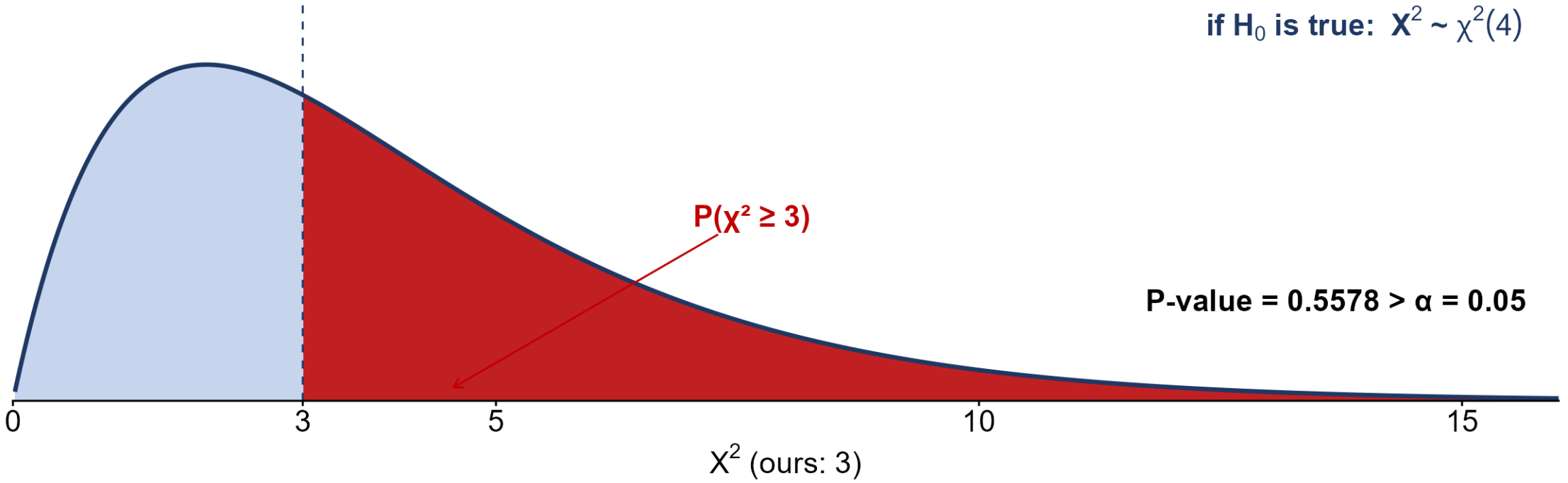
expected: $E = n\,p_{i,0} = 60{\times}0.2 = 12$ for every day

contribution: $(O-E)^2/E$; Monday: $(15-12)^2/12 = 0.75$

Day	Mon	Tue	Wed	Thu	Fri	Total
Observed O	15	12	9	9	15	**60**
Expected E	12	12	12	12	12	**60**
Contribution	0.75	0	0.75	0.75	0.75	**3**

$$X^2_{obs} = 0.75 + 0 + 0.75 + 0.75 + 0.75 = 3, \quad df = 5 - 1 = 4$$

The absences P-value



P-value = 0.5578 > $\alpha = 0.05$: fail to reject H_0; no evidence that the days differ

Example: streaming in the far West

A national study gives how many streaming services American families have. A random sample of 600 far-western families was asked the same. Does the far West differ? (OpenStax, Example 11.3)

services 0, 1, 2, 3, 4+: national 10%, 16%, 55%, 11%, 8%

far West observed 66, 119, 340, 60, 15; $n = 600,\ \alpha = 0.01$

(a) State H_0 and H_A.

(b) Find the expected counts and X^2_{obs}.

(c) Bracket the P-value. Decision? Which category misfits most?

(a) H_0**: the far West fits the national distribution;** H_A**: it does not**

(b) E **= 60, 96, 330, 66, 48;** $X^2_{obs} = 0.60 + 5.51 + 0.30 + 0.55 + 22.69 = 29.65$

(c) $df = 4$**; P-value < 0.0005 <** $\alpha = 0.01$**: reject; 4+ gives 22.69**

Practice: can we run the test?

Math faculty think 100 students' absences per term would be spread as below. A random survey of math courses counted actual absences. (OpenStax, Example 11.1)

absences 0-2, 3-5, 6-8, 9-11, 12+:

expected 50, 30, 12, 6, 2; observed 35, 40, 20, 1, 4

(a) Can we run a goodness-of-fit test on this table?

(b) If not, what do we do? What is the df then?

(a) no: the expected count for 12+ is 2, below 5

(b) combine 9-11 and 12+ into 9+ ($E = 8, O = 5$); $df = 4 - 1 = 3$

The test of independence

the question

are two categorical variables, measured on one sample, related?

the hypotheses

H_0**: the variables are independent;** H_A**: they are not (associated)**

the expected counts

$E = (\text{row total})(\text{column total})/n$ **for every cell**

the statistic and df

X^2_{obs} **adds a term for every cell;** $df = (r-1)(c-1)$**; upper tail**

Data: volunteers and hours (observed)

Type of volunteer	1-3 hours	4-6 hours	7-9 hours	Row total
Community college	111	96	48	**255**
Four-year college	96	133	61	**290**
Nonstudents	91	150	53	**294**
Column total	**298**	**379**	**162**	**839**

A sample of 839 adult volunteers, classified by type of volunteer and hours volunteered per week. (OpenStax, Example 11.6)

Example: volunteers and hours

Is the number of hours volunteered independent of the type of volunteer? (OpenStax, Example 11.6)

row totals 255, 290, 294; column totals 298, 379, 162; $n = 839$, $\alpha = 0.05$

(a) State H_0 and H_A.
(b) Find the table of expected counts.
(c) Find each cell's $(O - E)^2/E$ and X^2_{obs}.
(d) What is the df? Bracket the P-value. Decision?

(a) H_0: hours and type are independent; H_A: they are associated
(b), (c) on the next two slides: $X^2_{obs} = 12.99$
(d) $df = (3-1)(3-1) = 4$; 0.005 < P-value < 0.025 < $\alpha = 0.05$: reject H_0

Volunteers: expected counts

$E = (\text{row total})(\text{column total})/n$; community college, 1-3 hours: $255 \times 298/839 = 90.57$

Expected E	1-3 hours	4-6 hours	7-9 hours	Row total
Community college	90.57	115.19	49.24	**255**
Four-year college	103.00	131.00	56.00	**290**
Nonstudents	104.42	132.81	56.77	**294**
Column total	**298**	**379**	**162**	**839**

every $E \geq 5$; each row of E adds to its row total

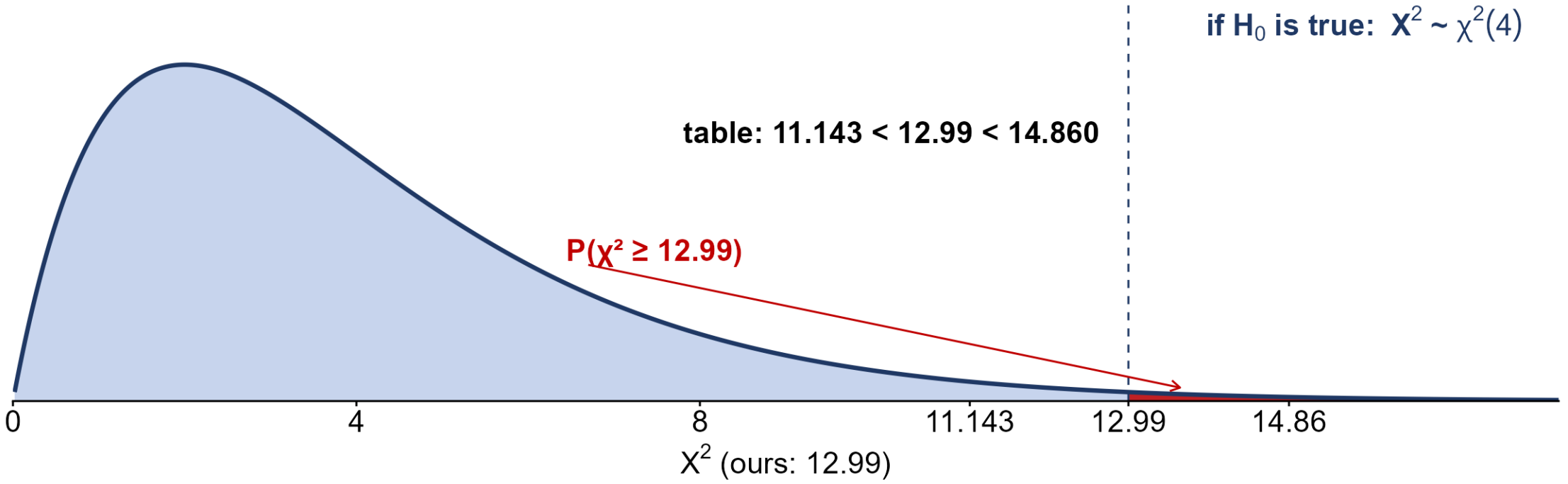
Volunteers: each cell's contribution

contribution $= (O - E)^2/E$; community college, 1-3 hours: $(111 - 90.57)^2/90.57 = 4.607$

Contribution	1-3 hours	4-6 hours	7-9 hours
Community college	4.607	3.197	0.031
Four-year college	0.476	0.030	0.447
Nonstudents	1.726	2.225	0.250

$$X^2_{obs} = 4.607 + 3.197 + 0.031 + 0.476 + 0.030 + 0.447 + 1.726 + 2.225 + 0.250 = 12.99$$

The volunteers P-value



0.005 < P-value < 0.025 < $\alpha = 0.05$: reject H_0; hours depend on the type of volunteer

Data: anxiety and need to succeed

Need / anxiety	High	Med-high	Medium	Med-low	Low	Total
High	35	42	53	15	10	**155**
Medium	18	48	63	33	31	**193**
Low	4	5	11	15	17	**52**
Total	**57**	**95**	**127**	**63**	**58**	**400**

A random sample of 400 De Anza College students took a test measuring their anxiety level and their need to succeed in school. (OpenStax, Example 11.7)

Practice: anxiety and need to succeed

De Anza College wants to know whether anxiety level and the need to succeed in school are related.

$n = 400$; **smallest expected count 7.41;** $\alpha = 0.05$

(a) If they are independent, how many high-need, high-anxiety students do we expect?
(b) How many low-need, med-low-anxiety students?
(c) State H_0 and H_A. What is the df?
(d) The 15 terms add to $X^2_{obs} = 48.42$. P-value? Decision?

(a) $155 \times 57/400 = 22.09$ **(b)** $52 \times 63/400 = 8.19$
(c) H_0**: independent;** H_A**: related;** $df = (3-1)(5-1) = 8$
(d) P-value < 0.0005 < $\alpha = 0.05$**: reject: they are related**

The test for homogeneity

the question

do two or more populations have the same distribution of one categorical variable?

the design

a separate random sample from each population: one row per population

the hypotheses

H_0**: the distributions are the same;** H_A**: they are not all the same**

the math

the same as independence: $E = (\text{row total})(\text{column total})/n$**,** $df = (r-1)(c-1)$

Example: living arrangements

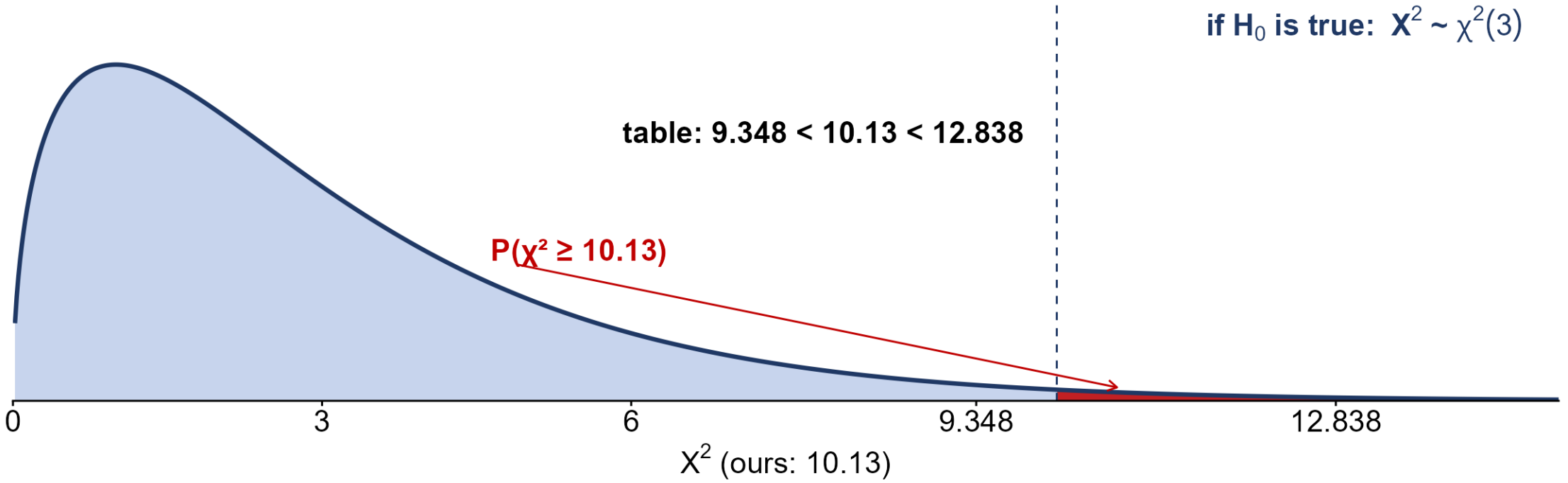
250 randomly selected men and 300 randomly selected women college students were asked where they live. Do men and women have the same distribution of living arrangements? (OpenStax, Example 11.8)

dormitory, apartment, with parents, other:
men 72, 84, 49, 45; women 91, 86, 88, 35; $\alpha = 0.05$

(a) Independence or homogeneity? State H_0 and H_A.
(b) Find E for men in a dormitory (column total 163, $n = 550$).
(c) The terms add to $X^2_{obs} = 10.13$. df? P-value? Decision?

(a) homogeneity: two samples, one variable; H_0: same distribution; H_A: not
(b) $E = 250 \times 163/550 = 74.09$
(c) $df = (2-1)(4-1) = 3$; **0.005 < P-value < 0.025 <** $\alpha = 0.05$: **reject**

The living-arrangements P-value



P-value = 0.0175 < $\alpha = 0.05$: reject; women live with parents more (0.293 vs 0.196)

Practice: voters and the earthquake

Before and after a recent earthquake, separate surveys asked voters which of three city-council candidates they planned to vote for. Did preferences change? (OpenStax, Example 11.9)

Perez, Chung, Stevens:

before 167, 128, 135; after 214, 197, 225; $\alpha = 0.05$

(a) Which test? State H_0 and H_A.

(b) Find E for before, Chung (row total 430, column total 325, $n = 1066$).

(c) The terms add to $X^2_{obs} = 3.26$. df? P-value? Decision?

(a) homogeneity; H_0: same distribution before and after; H_A: not

(b) $E = 430 \times 325/1066 = 131.10$: the smallest, so every $E \geq 5$

(c) $df = 2$; 0.15 < P-value < 0.2; P-value > $\alpha = 0.05$: fail to reject

Which chi-square test?

1 **Bags are said to be 20% of each of five colors; you count 300 candies.**

goodness of fit: one variable vs a claim; $E = 60$ per color, $df = 4$

2 **500 voters give their age group and their main news source.**

independence: one sample, two variables on each voter

3 **200 first-years and 200 seniors say how they get to campus.**

homogeneity: two separate samples, one variable; $df = (2-1)(4-1) = 3$

Summary: the three chi-square tests

goodness of fit

one sample, one variable vs a claim: $E = n\,p_{i,0}$, $df = k - 1$

independence

one sample, two variables: $E = (\text{row total})(\text{column total})/n$, $df = (r-1)(c-1)$

homogeneity

one sample per population, one variable: the same E and df as independence

all three

$X^2_{obs} = \sum(O-E)^2/E$; **upper tail; every** $E \geq 5$